

On Graph and Hypergraph Inference via Structural Equation Models

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Abstract—This article provides an overview of graph and hypergraph learning methods based on nodal observations as well as measurements arising from groups of nodes. We focus on a widely used class of approaches based on structural equation models (SEMs), in which each nodal observation is modeled as a linear combination of other nodal observations and, optionally, products of nodal observations or group-based measurements. These SEMs may also incorporate exogenous variables and noise. In conventional graph learning, the model reduces to a purely linear form by omitting product terms and group measurements, whereas their inclusion enables the recovery of higher-order interaction structures. We review several topology recovery problems, discuss their identifiability conditions, and summarize relevant references.

Index Terms—Directed graph learning, hypergraph learning, topology identification, structural equation models.

I. INTRODUCTION

Learning relational structure from multivariate observations is a fundamental problem in signal processing, machine learning, and network science, with applications including sensor networks, neuroscience, and biological systems. In many of these settings, dependencies among variables can be naturally modeled using graphs, and graph topology identification has therefore received considerable attention [1].

Structural equation models (SEMs) provide a flexible and interpretable framework for graph topology identification by explicitly modeling each nodal observation as a linear function of other nodal observations [2]–[5], possibly in the presence of noise and external inputs. SEMs have a long history in the inference of directed acyclic graphs (DAGs), particularly in the context of causal discovery and Bayesian networks [2]. Beyond acyclic structures, SEMs have also been proposed for the inference of general directed graphs, allowing for feedback loops and cyclic dependencies that frequently arise in networked dynamical systems. In this context, exogenous variables or node-specific external inputs have, for instance, been introduced in the model [5]. Without such exogenous variables, general directed and undirected graphs are difficult to infer, since the conventional signal matching criterion that works for DAGs becomes defective. To address this issue, a covariance matching criterion has been proposed in [6], [7].

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Under this principled framework for graph recovery, general SEMs can be recovered with limited prior knowledge.

While graphs capture pairwise interactions, many real-world systems exhibit higher-order interactions involving more than two nodes simultaneously. This has motivated the use of hypergraphs [8], [9] and simplicial complexes [10], [11]. These developments have been accompanied by corresponding extensions of SEMs [12]–[14], which we refer to as hypergraph SEMs. Such SEMs include additional higher-order terms that are either constructed from products of nodal observations or arise from new higher-level observations, such as flows in transportation systems, tomographical measurements, or the number of papers produced by a group of authors. Research on hypergraph recovery using SEMs is still in its infancy, but efficient algorithms have already been studied. In particular, [12] presents identifiability conditions based on exogenous variables, [13] introduces simplicial complex constraints, and [14] focuses on the reconstruction of directed acyclic hypergraphs.

The goal of this paper is to provide a unified overview of SEM-based methods for graph and hypergraph inference, spanning directed and undirected structures as well as hypergraph and simplicial-complex representations. Rather than proposing new algorithms, we synthesize existing approaches, highlight their connections, and clarify modeling assumptions and structural constraints.

II. PRELIMINARIES

This section introduces the concepts required for the proposed unifying framework. We first review basic notions of graphs and graph signals and then extend these concepts to hypergraphs and hypergraph signals. We subsequently combine topology and signals to define both the conventional SEM and the hypergraph SEM. These SEMs serve as the foundation for learning graphs and hypergraphs in the subsequent sections.

A. Graphs and Graph Signals

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed (weighted) graph, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ denotes the set of nodes, and $\mathcal{E} = \{e_1, e_2, \dots, e_M\}$ denotes the set of edges. Each edge is defined by the triplet $e_m = (s_m, t_m, w_m)$, where $s_m \in \mathcal{V}$ is the source node, $t_m \in \mathcal{V} \setminus s_m$ is the target node, and $w_m \in \mathbb{R}$ represents the edge weight. Self-loops are excluded. A path in a graph is a succession of edges such that the target of any edge is the

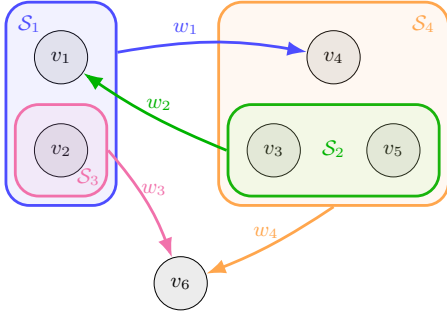


Fig. 1: A weighted directed hypergraph of order $P = 3$.

source of the subsequent edge. A cycle is a path for which the source of the first edge coincides with the target of the last edge.

A graph (nodal) signal is defined as a vector $\mathbf{x} \in \mathbb{R}^N$ with one entry per node. Such signals may, for instance, arise from sensor measurements, in which case the sensors correspond to the graph nodes.

B. Hypergraphs and Hypergraph Signals

A directed (weighted) hypergraph is a natural extension of the previously defined directed (weighted) graph [15]–[19]. It can be represented as $\mathcal{H} = (\mathcal{V}, \mathcal{H})$, where $\mathcal{V}$ denotes the set of nodes and $\mathcal{H} = \{h_1, \dots, h_K\}$ is the set of hyperedges. Each hyperedge is defined by the triplet $h_k = (\mathcal{S}_k, t_k, w_k)$, where the source is a subset of nodes $\mathcal{S}_k \subset \mathcal{V}$, the target is a single node¹ $t_k \in \mathcal{V} \setminus \mathcal{S}_k$, and $w_k \in \mathbb{R}$ is the weight associated with the hyperedge. Self-loops are again avoided by excluding the source set from the set of potential target nodes. The order P of a hypergraph is defined as the maximum cardinality of a source set $\mathcal{S}_k$. Note that this definition of hyperedges includes conventional edges, for which both the source and the target consist of single nodes. See Fig. 1 for an example of a hypergraph with $P = 3$.

A hypergraph signal is defined as a vector $\mathbf{y} \in \mathbb{R}^L$, with one entry for each potential source set $\mathcal{V}_l \subset \mathcal{V}$, where $l = 1, 2, \dots, L$. For a hypergraph of order P , the number of possible source sets is $L = \sum_{p=1}^P L_p$, with $L_p = \binom{N}{p}$ denoting the number of candidate source sets of size p . We order the entries of $\mathbf{y}$ as $\mathbf{y} = [\mathbf{y}_1^\top, \mathbf{y}_2^\top, \dots, \mathbf{y}_P^\top]^\top$, where $\mathbf{y}_p \in \mathbb{R}^{L_p}$ is the level- p hypergraph signal that collects the signal values associated with all potential groups of nodes of size p . Note that $\mathbf{y}_1 \in \mathbb{R}^N$ contains the nodal signal values, i.e., $\mathbf{y}_1 = \mathbf{x}$.

Higher-level hypergraph signals $\{\mathbf{y}_p\}_{p=2}^P$ may arise naturally from interactions among the nodes in a source set, with practical examples including tomographical measurements, flows in transportation systems, or word frequencies in papers produced by a group of authors. Alternatively, the signals $\{\mathbf{y}_p\}_{p=2}^P$ can be constructed from nodal signals, for instance via products or other permutation-invariant functions [12]. As an example, the level- p hypergraph signal value $[\mathbf{y}]_l$ associated

with the set $\mathcal{V}_l = \{v_{n_1}, v_{n_2}, \dots, v_{n_p}\}$ can be defined as $[\mathbf{y}]_l = x_{n_1} x_{n_2} \dots x_{n_p}$.

C. Structural Equation Models

SEMs connect signals to a topology. For graphs and graph signals, the SEM can be written as [2]–[5]

$$\mathbf{x} = \mathbf{W}_1 \mathbf{x} + \mathbf{v} + \mathbf{n}, \quad (1)$$

where $\mathbf{W}_1 \in \mathbb{R}^{N \times N}$ is the weighted adjacency matrix that contains the edge weights w_m . More specifically, if there exists an edge e_m with $m \in \{1, 2, \dots, M\}$ such that the edge target is $t_m = v_i$ and the edge source is $s_m = v_j$, then $[\mathbf{W}_1]_{i,j} = w_m$; otherwise, $[\mathbf{W}_1]_{i,j} = 0$. Further, $\mathbf{v}$ denotes an exogenous input, and $\mathbf{n}$ is zero-mean additive noise.

To extend this conventional SEM to the hypergraph setting, we express the nodal signal $\mathbf{x}$ as a function of the hypergraph signal $\mathbf{y}$ through the hypergraph topology. Specifically, we define the hypergraph SEM as [12]–[14]

$$\mathbf{x} = \mathbf{W} \mathbf{y} + \mathbf{v} + \mathbf{n}, \quad (2)$$

where $\mathbf{W} \in \mathbb{R}^{N \times L}$ is the weight matrix containing the hyperedge weights w_k . More precisely, if there exists a hyperedge h_k with $k \in \{1, 2, \dots, K\}$ such that the hyperedge target is $t_k = v_i$ and the hyperedge source set is $\mathcal{S}_k = \mathcal{V}_j$, then $[\mathbf{W}]_{i,j} = w_k$; otherwise, $[\mathbf{W}]_{i,j} = 0$. Again, the model includes an exogenous input $\mathbf{v}$ and zero-mean additive noise $\mathbf{n}$.

Since $\mathbf{y}_1 = \mathbf{x}$, the hypergraph SEM in (2) can equivalently be rewritten as

$$\mathbf{x} = \mathbf{W}_1 \mathbf{x} + \sum_{p=2}^P \mathbf{W}_p \mathbf{y}_p + \mathbf{v} + \mathbf{n}, \quad (3)$$

where $\mathbf{W}_1 \in \mathbb{R}^{N \times N}$ represents the pairwise edges in the hypergraph, and $\mathbf{W}_p \in \mathbb{R}^{N \times L_p}$ characterizes how signals defined on p -tuples of nodes influence the nodal signal. Note that $\mathbf{W}$ can be written as $\mathbf{W} = [\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_P]$. In the special case where higher-level hypergraph signals are constructed as products of nodal observations, the hypergraph SEM is also referred to as the Volterra SEM [12].

III. SEM-BASED INFERENCE FOR GRAPHS

We first focus on the conventional SEM in (1). Given T measurements $\mathbf{x}(t)$, $t = 1, 2, \dots, T$, and their corresponding exogenous observations $\mathbf{v}(t)$, a popular approach to recover the weighted adjacency matrix $\mathbf{W}_1$ is to solve the signal matching problem

$$\min_{\mathbf{W}_1} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W}_1 \mathbf{x}(t) - \mathbf{v}(t)\|_2^2, \quad (4)$$

where sparsity can be promoted via an ℓ_1 -regularizer $\|\mathbf{W}_1\|_1$. In a noiseless setting, identifiability conditions for this problem have been presented in [5], showing that the exogenous variables must be sufficiently exciting to guarantee identifiability.

With zero-mean white Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ but without exogenous variables, the SEM in (1) reduces to

$$\mathbf{x} = \mathbf{W}_1 \mathbf{x} + \mathbf{n} \Rightarrow \mathbf{x} = (\mathbf{I} - \mathbf{W}_1)^{-1} \mathbf{n}, \quad (5)$$

which implies that $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_x)$ with $\mathbf{R}_x = \sigma^2 (\mathbf{I} - \mathbf{W}_1)^{-1} (\mathbf{I} - \mathbf{W}_1^\top)^{-1}$. In this setting, a generic (un)directed

¹Hypergraphs in which targets are sets of nodes also exist but, to keep the exposition simple, they are not considered in this work.

graph is not identifiable via the signal matching optimization in (4) [6], [7]. Interestingly, if the optimization is augmented with DAG constraints, $\mathbf{W}_1$ becomes identifiable [4]. For this reason, we first discuss SEM-based signal matching approaches for learning DAGs and then present a SEM-based covariance matching approach for learning general graphs.

Two important properties of adjacency matrices corresponding to DAGs are that i) the determinant of $\mathbf{W}_1$ is zero, and ii) with a suitable ordering of the nodes, $\mathbf{W}_1$ is strictly upper triangular. As a consequence, in the absence of exogenous variables, the signal matching problem in (4) maximizes the likelihood of $\mathbf{x}$ in (5) under zero-mean white Gaussian noise. The DAG learning problem can therefore be formulated as

$$\begin{aligned} \min_{\mathbf{W}_1} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W}_1 \mathbf{x}(t)\|_2^2 \\ \text{s.t. } \mathbf{W}_1 \text{ represents a DAG.} \end{aligned} \quad (6)$$

Early approaches to DAG recovery relied on combinatorial search methods [4]. These approaches motivated more efficient algorithms that replace the explicit DAG constraint with a (non-convex) continuous regularizer, enabling the use of continuous optimization techniques to estimate $\mathbf{W}_1$ [20]–[22]. Owing to its favorable properties, we adopt the regularizer proposed in DAGMA [21],

$$f(\mathbf{W}_1) = N \log s - \log(\det(s\mathbf{I}_N - \mathbf{W}_1 \circ \mathbf{W}_1)), \quad (7)$$

where $\circ$ denotes the element-wise product and $s \in \mathbb{R}^+$ is a freely chosen parameter. A relaxed version of (6) can then be stated as

$$\min_{\mathbf{W}_1} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W}_1 \mathbf{x}(t)\|_2^2 + \nu f(\mathbf{W}_1), \quad (8)$$

where ν is the acyclicity regularization weight. Following DAGMA [21], we gradually decrease s and increase ν during optimization to enforce acyclicity and obtain a DAG.

We now turn to the case where the underlying graph is not a DAG. Assuming again zero-mean white Gaussian noise and no exogenous variables, the exact maximum likelihood problem can be formulated. Although this problem deserves further study, thus far we approximated it with a covariance matching approach [6], [7]. Defining the sample covariance matrix as $\hat{\mathbf{R}}_x = \frac{1}{T} \sum_{t=1}^T \mathbf{x}(t) \mathbf{x}^\top(t)$ and recalling that $\mathbf{R}_x = \sigma^2(\mathbf{I} - \mathbf{W}_1)^{-1}(\mathbf{I} - \mathbf{W}_1^\top)^{-1}$, we can formulate the covariance matching problem as

$$\min_{\mathbf{W}_1} \|\hat{\mathbf{R}}_x - \sigma^2(\mathbf{I} - \mathbf{W}_1)^{-1}(\mathbf{I} - \mathbf{W}_1^\top)^{-1}\|_F^2. \quad (9)$$

For general undirected and directed graphs, this problem can be further simplified to a least-squares problem in a binary vector (for undirected graphs) or an orthogonal matrix (for directed graphs); see [7] for details. As in previous settings, sparsity can be promoted via ℓ_1 -norm regularization.

IV. FROM GRAPHS TO HYPERGRAPHS

The signal matching problem can be extended to the hypergraph SEM in (2) or (3), yielding [12]–[14]

$$\min_{\mathbf{W}} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W} \mathbf{y}(t) - \mathbf{v}(t)\|_2^2, \quad (10)$$

or, using the block decomposition of $\mathbf{W}$, we can write the alternative formulation

$$\min_{\{\mathbf{W}_p\}} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W}_1 \mathbf{x}(t) - \sum_{p=2}^P \mathbf{W}_p \mathbf{y}_p(t) - \mathbf{v}(t)\|_2^2. \quad (11)$$

In a noiseless setting, identifiability conditions analogous to those in [5] have been presented in [12]. Although [12] focuses on the Volterra SEM, the proposed conditions also extend to more general hypergraph SEMs with hyperedge signals that are not constructed from the nodal signals.

In the absence of exogenous variables, the identifiability of (10) remains an open problem. Compared to (4), the additional term in (11) appears to facilitate the recovery of $\mathbf{W}$ even without exogenous inputs, although a formal proof is still lacking. Nevertheless, imposing restrictions on the class of admissible hypergraphs can only enhance identifiability. Two important examples analyzed in literature are discussed next.

A. Directed Acyclic Hypergraph Recovery

The first class of hypergraphs we consider is that of directed acyclic hypergraphs (DAHs) [14], [18]. We define a path in a hypergraph as an alternating sequence of hyperedges and nodes, where each hyperedge is followed by one of its target nodes, and each node is followed by a hyperedge whose source set contains that node. A cycle is then a path whose last node is contained in the source set of the first hyperedge. A DAH is a directed hypergraph without any cycles.

In the absence of exogenous variables, the DAH recovery problem can be formulated as [14]

$$\begin{aligned} \min_{\mathbf{W}} \sum_{t=1}^T \|\mathbf{x}(t) - \mathbf{W} \mathbf{y}(t)\|_2^2 \\ \text{s.t. } \mathbf{W} \text{ represents a DAH.} \end{aligned} \quad (12)$$

The DAH constraint cannot be directly replaced by a continuous regularizer, as in the DAG case. To address this difficulty, the directed hypergraph associated with $\mathbf{W}$ can be represented by a conventional (lifted) directed graph. See, for instance, [8] for several graph representations of a hypergraph, albeit limited to the undirected case. Among these, we focus on: a) the directed version of the star expansion (also known as the functional dependency graph [23]) and b) the line graph.

Star expansion: The star expansion is a bipartite graph composed of two disjoint node sets. One set contains the N regular nodes $\mathcal{V}$, while the other contains nodes corresponding to all possible source sets $\mathcal{V}_l$ of size up to P . This set is referred to as the *virtual* node set $\mathcal{U} = \{u_1, u_2, \dots, u_L\}$, with each virtual node u_l associated with a specific group of regular nodes $\mathcal{V}_l$. The adjacency matrix of the star expansion is

$$\mathbf{W}_* = \begin{bmatrix} \mathbf{0} & \mathbf{W} \\ \mathbf{S} & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{(N+L) \times (N+L)},$$

where $\mathbf{S} \in \{0, 1\}^{L \times N}$ indicates which regular nodes belong to which virtual nodes, i.e., $[\mathbf{S}]_{i,j} = 1$ if node v_j is an element of $\mathcal{V}_i$ associated with the i th virtual node u_i , and it is zero otherwise. As a result, the star expansion is a bipartite graph with edges in both directions. A virtual node u_l points to a regular node v_n if there exists a hyperedge h_k with $\mathcal{S}_k = \mathcal{V}_l$

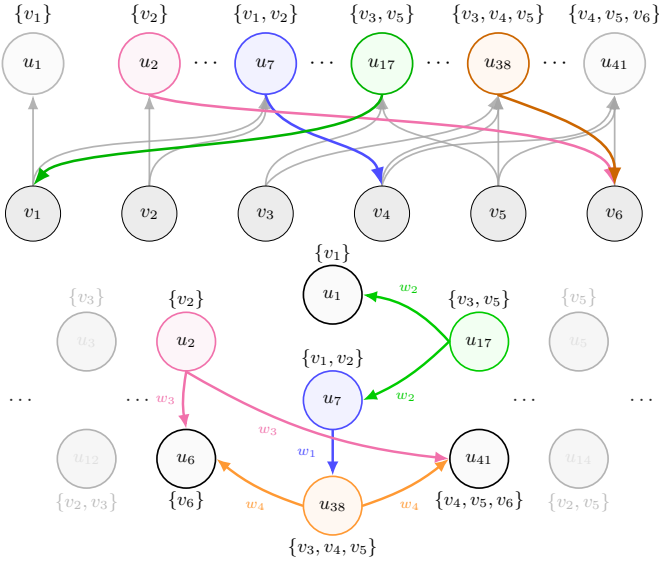


Fig. 2: The star expansion (top) and line graph (bottom) related to the hypergraph of Fig. 1.

and $t_k = v_n$, in which case the weight of that edge is w_k . Additionally, a regular node v_n points to a virtual node u_l if $v_n \in \mathcal{V}_l$. The weight of such an edge is arbitrarily set to 1. The star expansion of the hypergraph of Fig. 1 is depicted on the top of Fig. 2.

Line graph: The line graph is a graph defined on the L virtual nodes $\mathcal{U}$. Its adjacency matrix is

$$\mathbf{W}_- = \mathbf{S}\mathbf{W} \in \mathbb{R}^{L \times L}.$$

Equivalently, an edge in the line graph points from the j th virtual node u_j to the i th virtual node u_i if there exists a hyperedge h_k with source set $\mathcal{S}_k = \mathcal{V}_j$, represented by u_j , and target node t_k contained in $\mathcal{V}_i$, represented by u_i . The weight of that edge is w_k . The line graph of the hypergraph of Fig. 1 is shown at the bottom of Fig. 2.

Having introduced these two directed graph representations of a directed hypergraph, the following claim holds: *a directed hypergraph is a DAH if and only if the star expansion or the line graph is a DAG*. Accordingly, we can replace the DAH constraint in (12) by a DAG regularizer based on a graph with adjacency matrix $\mathbf{W}_*$ or $\mathbf{W}_-$. For the star expansion, this yields [14]

$$\begin{aligned} f(\mathbf{W}_*) &= (N + L) \log s - \log(\det(s\mathbf{I}_{L+N} - \mathbf{W}_* \circ \mathbf{W}_*)) \\ &= -\log \det(\mathbf{I}_L - s^{-2}(\mathbf{S} \circ \mathbf{S})(\mathbf{W} \circ \mathbf{W})), \end{aligned} \quad (13)$$

whereas for the line graph we obtain

$$\begin{aligned} f(\mathbf{W}_-) &= L \log s - \log(\det(s\mathbf{I}_L - \mathbf{W}_- \circ \mathbf{W}_-)) \\ &= -\log \det(\mathbf{I}_L - s^{-1}(\mathbf{S}\mathbf{W}) \circ (\mathbf{S}\mathbf{W})). \end{aligned} \quad (14)$$

Both functions (13) and (14) enforce that all powers of a suitably defined nonnegative matrix, with the same sparsity pattern as $\mathbf{S}\mathbf{W}$, have zero diagonal. In (13), this matrix is $(\mathbf{S} \circ \mathbf{S})(\mathbf{W} \circ \mathbf{W})$, whereas in (14) it is $(\mathbf{S}\mathbf{W}) \circ (\mathbf{S}\mathbf{W})$.

B. Symmetry and Inclusion Constraints

Another way to restrict the class of hypergraphs is to impose symmetry and inclusion constraints, which are particularly important for so-called simplicial complexes [13].

- **Symmetry** means that the ordering of the nodes in the union of a source set and its target node is irrelevant.
- **Inclusion**, in our setting, means that whenever a source set (virtual node) is active and points to a specific target node, all possible subsets of this source set should also be active and point to that same target node.

To provide clear examples, we restrict attention to hypergraphs of order $P = 2$. In this case, the hypergraph SEM in (3) is fully characterized by $\mathbf{W}_1$ and $\mathbf{W}_2$. To simplify notation, we denote the entry in $\mathbf{W}_2$ linking node n to the pair of nodes $\{i, j\}$ by $[\mathbf{W}_2]_{n, \{i, j\}}$. We can then introduce the following symmetry and inclusion constraints.

Edge symmetry: This constraint enforces conventional pairwise edges to be undirected. Specifically, it requires that $[\mathbf{W}_1]_{i, j} = [\mathbf{W}_1]_{j, i}$ for all $i, j = 1, 2, \dots, N$ with $i < j$.

Triangle symmetry: Similarly, symmetry within a triangle can be enforced by constraining all potential hyperedges in that triangle to have the same weight. Concretely, this leads to the constraints $[\mathbf{W}_2]_{n, \{i, j\}} = [\mathbf{W}_2]_{n, \{i, n\}} = [\mathbf{W}_2]_{j, \{i, n\}}$, for all $n, i, j = 1, 2, \dots, N$ with $n < i < j$.

Inclusion: Based on the earlier definition, inclusion means that a hyperedge between a source pair and a target node can only exist if there is an edge between each node of the source pair and the target. In terms of $\mathbf{W}_1$ and $\mathbf{W}_2$, this implies that $[\mathbf{W}_2]_{n, \{i, j\}} \neq 0$ requires $[\mathbf{W}_1]_{n, i} \neq 0$ and $[\mathbf{W}_1]_{n, j} \neq 0$. This constraint can be relaxed by introducing a group sparsity regularizer of the form

$$\sum_{n < i < j}^N \left\| \begin{bmatrix} [\mathbf{W}_2]_{n, \{i, j\}}, [\mathbf{W}_1]_{n, i}, [\mathbf{W}_1]_{n, j} \end{bmatrix}^\top \right\|_2.$$

In practice, this inclusion constraint is often combined with edge and triangle symmetry, resulting in a simplicial complex structure. Such a setting has for instance been considered in [13].

V. CONCLUSIONS

This article reviewed graph and hypergraph topology learning methods based on nodal and/or group-level observations, with an emphasis on approaches grounded in structural equation models. By extending conventional linear SEM formulations to include product terms or group measurements, these models enable the identification of higher-order interaction structures. We discussed several topology recovery problems within this framework and highlighted key identifiability conditions governing their successful application. The presented overview and references aim to clarify the relationships among existing methods and to provide guidance for the development and analysis of future graph and hypergraph learning algorithms.

REFERENCES

- [1] G. Mateos, S. Segarra, A. G. Marques, and A. Ribeiro, “Connecting the dots: Identifying network structure via graph signal processing,” *IEEE Signal Process. Mag.*, vol. 36, no. 3, pp. 16–43, 2019.
- [2] J. Pearl, “Graphs, causality, and structural equation models,” *Sociol. Methods Res.*, vol. 27, no. 2, pp. 226–284, 1998.
- [3] S. Shimizu, P. O. Hoyer, A. Hyvärinen, A. Kerminen, and M. Jordan, “A linear non-Gaussian acyclic model for causal discovery,” *J. Mach. Learn. Res.*, vol. 7, no. 10, 2006.
- [4] J. Peters and P. Bühlmann, “Identifiability of Gaussian structural equation models with equal error variances,” *Biometrika*, vol. 101, no. 1, pp. 219–228, 2013.
- [5] J. A. Bazerque, B. Baingana, and G. B. Giannakis, “Identifiability of sparse structural equation models for directed and cyclic networks,” in *IEEE Global Conf. Signal Inf. Process. (GlobalSIP)*, 2013, pp. 839–842.
- [6] Y. Han, A. Natali, and G. Leus, “Graph topology identification based on covariance matching,” in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2025.
- [7] Y. Han, R. T. Rajan, and G. Leus, “Graph topology identification based on covariance matching,” *arXiv preprint arXiv:2601.15999*, 2026.
- [8] M. T. Schaub, Y. Zhu, J. B. Seby, T. M. Roddenberry, and S. Segarra, “Signal processing on higher-order networks: Livin’ on the edge... and beyond,” *Signal Process.*, vol. 187, pp. 108149, 2021.
- [9] S. Zhang, Z. Ding, and S. Cui, “Introducing hypergraph signal processing: Theoretical foundation and practical applications,” *IEEE Internet Things J.*, vol. 7, no. 1, pp. 639–660, Jan. 2020.
- [10] S. Barbarossa and S. Sardellitti, “Topological signal processing over simplicial complexes,” *IEEE Trans. Signal Process.*, vol. 68, pp. 2992–3007, 2020.
- [11] E. Isufi, G. Leus, B. Beferull-Lozano, S. Barbarossa, and P. Di Lorenzo, “Topological signal processing and learning: Recent advances and future challenges,” *Signal Process.*, p. 109930, 2025.
- [12] Q. Yang, M. Coutino, G. Leus, and G. B. Giannakis, “Autoregressive graph Volterra models and applications,” *EURASIP J. Adv. Signal Process.*, vol. 2023, no. 4, 2023.
- [13] A. Buciualea, E. Isufi, G. Leus, and A. G. Marques, “Learning graphs and simplicial complexes from data,” in *IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*. IEEE, 2024, pp. 9861–9865.
- [14] G. Leus, Y. Han, E. Isufi, and A. G. Marques, “Learning directed acyclic hypergraphs from structural equation model topological data,” in *Asilomar Conf. Signals, Syst., Comput.*, 2025, pp. 1–5.
- [15] G. Gallo, G. Longo, S. Pallottino, and S. Nguyen, “Directed hypergraphs and applications,” *Discrete Appl. Math.*, vol. 42, pp. 177–201, 1993.
- [16] A. R. Berg, B. Jackson, and T. Jordán, “Edge splitting and connectivity augmentation in directed hypergraphs,” *Discrete Math.*, vol. 273, no. 1–3, pp. 71–84, 2003.
- [17] A. Frank, T. Király, and Z. Király, “On the orientation of graphs and hypergraphs,” *Discrete Appl. Math.*, vol. 131, no. 2, pp. 385–400, Sept. 2003.
- [18] M. Thakur and R. Tripathi, “Linear connectivity problems in directed hypergraphs,” *Theor. Comput. Sci.*, vol. 410, no. 27, pp. 2592–2618, 2009.
- [19] G. Ausiello and L. Laura, “Directed hypergraphs: Introduction and fundamental algorithms—a survey,” *Theor. Comput. Sci.*, vol. 658, pp. 293–306, 2017.
- [20] X. Zheng, B. Aragam, P. Ravikumar, and E. P. Xing, “DAGs with NO TEARS: Continuous optimization for structure learning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2018.
- [21] K. Bello, B. Aragam, and P. Ravikumar, “DAGMA: Learning DAGs via M-matrices and a log-determinant acyclicity characterization,” in *Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 8226–8239.
- [22] S. S. Saboksayr, G. Mateos, and M. Tepper, “Block successive convex approximation for concomitant linear dag estimation,” in *IEEE Sensor Array Multichannel Signal Process. Workshop (SAM)*, 2024, pp. 1–5.
- [23] G. Ausiello, A. D’Atri, and D. Saccà, “Graph algorithms for functional dependency manipulation,” *J. ACM*, vol. 30, pp. 752–766, 1983.